

Optimization Lecture 3+4

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Outline

Convex functions

Operations that preserve convexity

Perspective and conjugate

Quasiconvexity

Summary

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Convex functions

Operations that preserve convexity

Perspective and conjugate

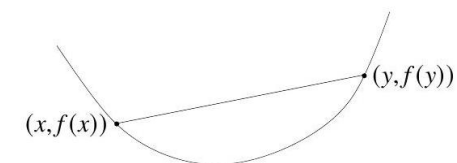
Quasiconvexity

Summary

Definition

- $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is convex if **dom** f is a convex set and for all $x, y \in \mathbf{dom} f, 0 \leq \theta \leq 1$,

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$



- f is concave if $-f$ is convex
- f is **strictly** convex if **dom** f is convex and for $x, y \in \mathbf{dom} f, x \neq y, 0 < \theta < 1$,
$$f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y)$$

can be generated to other spaces.

Restriction of a convex function to a line

- ▶ $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is convex if and only if the function $g : \mathbf{R} \rightarrow \mathbf{R}$,

$$g(t) = f(x + tv), \quad \text{dom } g = \{t \mid x + tv \in \text{dom } f\}$$

is convex in t for any $x \in \text{dom } f, v \in \mathbf{R}^n$

can be used to check convexity of f by checking convexity of functions of one variable

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Proof ($m = 1$)

- ▶ Necessary: For any $t \in (0, 1)$, and $x, y \in \text{dom } f$. By def:

$$f(x + t(y - x)) \leq (1 - t)f(x) + tf(y)$$

$$f(y) \geq f(x) + \frac{f(x + t(y - x)) - f(x)}{t}$$

Let $t \rightarrow 0$, we have $f(y) \leq f(x) + f'(x)(y - x)$

- ▶ Sufficiency: Let $z = \theta x + (1 - \theta)y$:

$$f(x) \geq f(z) + f'(z)(x - z)$$

and

$$f(y) \geq f(z) + f'(z)(y - z)$$

$\Rightarrow ??$

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First-order condition

- ▶ f is **differentiable** if $\text{dom } f$ is open and the gradient

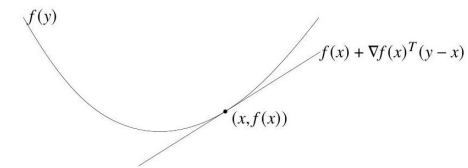
$$\nabla f(x) = \left(\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_n} \right)^T \in \mathbf{R}^n$$

exists at each $x \in \text{dom } f$

- ▶ **1st-order condition**: differentiable f with convex domain is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y \in \text{dom } f$$

- ▶ first order Taylor approximation of convex f is a **global underestimator** of f



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Second-order conditions

- ▶ f is **twice differentiable** if $\text{dom } f$ is open and the Hessian $\nabla^2 f(x) \in \mathbf{S}^n$,

$$\nabla^2 f(x)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}, \quad i, j = 1, \dots, n$$

exists at each $x \in \text{dom } f$

- ▶ **2nd-order conditions**: for twice differentiable f with convex domain
 - f is convex if and only if $\nabla^2 f(x) \geq 0$ for all $x \in \text{dom } f$
 - if $\nabla^2 f(x) > 0$ for all $x \in \text{dom } f$, then f is strictly convex

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Extended-value extension

- ▶ suppose f is convex on $\mathbf{R}^n$, with domain $\mathbf{dom} f$
- ▶ its extended-value extension $\tilde{f}$ is function $\tilde{f} : \mathbf{R}^n \rightarrow \mathbf{R} \cup \{\infty\}$

$$\tilde{f}(x) = \begin{cases} f(x) & x \in \mathbf{dom} f \\ \infty & x \notin \mathbf{dom} f \end{cases}$$

- ▶ often simplifies notation; for example, the condition

$$0 \leq \theta \leq 1 \implies \tilde{f}(\theta x + (1 - \theta)y) \leq \theta \tilde{f}(x) + (1 - \theta) \tilde{f}(y)$$

(as an inequality in $\mathbf{R} \cup \{\infty\}$), means the same as the two conditions

- $\mathbf{dom} f$ is convex
-

$$x, y \in \mathbf{dom} f, 0 \leq \theta \leq 1 \implies$$

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

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Examples on $\mathbf{R}^n$

convex functions:

- ▶ affine functions: $f(x) = a^T x + b$
- ▶ any norm, e.g., the ℓ_p norms
- ▶ $\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{1/p}$ for $p \geq 1$
- ▶ $\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$
- ▶ sum of squares: $\|x\|_2^2 = x_1^2 + \dots + x_n^2$
- ▶ max function: $\max(x) = \max\{x_1, x_2, \dots, x_n\}$
- ▶ softmax or log-sum-exp function: $\log(\exp x_1 + \dots + \exp x_n)$ (ex in class)

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Examples on $\mathbf{R}$

convex functions:

- ▶ affine: $ax + b$ on $\mathbf{R}$, for any $a, b \in \mathbf{R}$
- ▶ exponential: e^{ax} , for any $a \in \mathbf{R}$
- ▶ powers: x^α on $\mathbf{R}_{++}$, for $\alpha \geq 1$ or $\alpha \leq 0$
- ▶ powers of absolute value: $|x|^p$ on $\mathbf{R}$, for $p \geq 1$
- ▶ positive part (relu): $\max\{0, x\}$

concave functions:

- ▶ affine: $ax + b$ on $\mathbf{R}$, for any $a, b \in \mathbf{R}$ (by def)
- ▶ powers: x^α on $\mathbf{R}_{++}$, for $0 \leq \alpha \leq 1$
- ▶ logarithm: $\log x$ on $\mathbf{R}_{++}$
- ▶ entropy: $-x \log x$ on $\mathbf{R}_{++}$
- ▶ negative part: $\min\{0, x\}$ (by def)

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Examples on $\mathbf{R}^{m \times n}$

- ▶ $X \in \mathbf{R}^{m \times n}$ ($m \times n$ matrices) is the variable
- ▶ general affine function has form

$$f(X) = \text{tr}(A^T X) + b = \sum_{i=1}^m \sum_{j=1}^n A_{ij} X_{ij} + b$$

for some $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}$

- ▶ spectral norm (maximum singular value) is convex

$$f(X) = \|X\|_2 = \sigma_{\max}(X) = \left(\lambda_{\max}(X^T X) \right)^{1/2}$$

- ▶ log-determinant: for $X \in \mathbf{S}_{++}^n$, $f(X) = \log \det X$ is concave

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Example

- ▶ $f : \mathbf{S}^n \rightarrow \mathbf{R}$ with $f(X) = \log \det X$, $\text{dom } f = \mathbf{S}_{++}^n$
- ▶ consider line in $\mathbf{S}^n$ given by $X + tV$, $X \in \mathbf{S}_{++}^n$, $V \in \mathbf{S}^n$, $t \in \mathbf{R}$

$$\begin{aligned} g(t) &= \log \det(X + tV) \\ &= \log \det \left(X^{1/2} \left(I + tX^{-1/2} V X^{-1/2} \right) X^{1/2} \right) \\ &= \log \det X + \log \det \left(I + tX^{-1/2} V X^{-1/2} \right) \\ &= \log \det X + \sum_{i=1}^n \log(1 + t\lambda_i) \end{aligned}$$

where λ_i are the eigenvalues of $X^{-1/2} V X^{-1/2}$

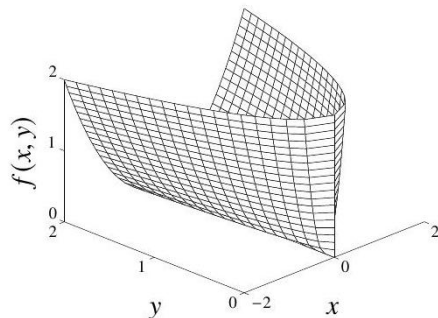
- g is concave in t (for any choice of $X \in \mathbf{S}_{++}^n, V \in \mathbf{S}^n$); hence f is concave

Examples (continued)

- ▶ **quadratic-over-linear:** $f(x, y) = x^2/y, y > 0$

$$\nabla^2 f(x, y) = \frac{2}{y^3} \begin{bmatrix} y \\ -x \end{bmatrix} \begin{bmatrix} y \\ -x \end{bmatrix}^T \geq 0$$

convex for $y > 0$



Examples

- **quadratic function:** $f(x) = (1/2)x^T Px + q^T x + r$ (with $P \in \mathbb{S}^n$)

$$\nabla f(x) = P x + q, \quad \nabla^2 f(x) = P$$

convex if $P \geq 0$ (concave if $P \leq 0$)

- ▶ **least-squares objective:** $f(x) = \|Ax - b\|_2^2$

$$\nabla f(x) = 2A^T(Ax - b), \quad \nabla^2 f(x) = 2A^T A$$

convex (for any A)

More examples

- **log-sum-exp:** $f(x) = \log \sum_{k=1}^n \exp x_k$ is convex

$$\nabla^2 f(x) = \frac{1}{\mathbf{1}^T \mathbf{z}} \mathbf{diag}(\mathbf{z}) - \frac{1}{(\mathbf{1}^T \mathbf{z})^2} \mathbf{z} \mathbf{z}^T \quad (\mathbf{z}_k = \exp x_k)$$

- ▶ to show $\nabla^2 f(x) \succeq 0$, we must verify that $v^T \nabla^2 f(x) v \geq 0$ for all v :

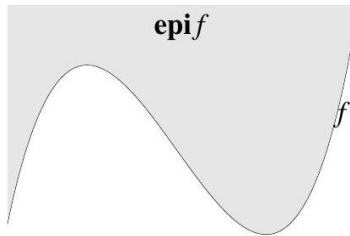
$$v^T \nabla^2 f(x) v = \frac{(\sum_k z_k v_k^2)(\sum_k z_k) - (\sum_k v_k z_k)^2}{(\sum_k z_k)^2} \geq 0$$

since $(\sum_k v_k z_k)^2 \leq (\sum_k z_k v_k^2) (\sum_k z_k)$ (from Cauchy-Schwarz inequality)

- ▶ **geometric mean:** $f(x) = (\prod_{k=1}^n x_k)^{1/n}$ on $\mathbf{R}_{++}^n$ is concave (similar proof as above)

Epigraph and sublevel set

- ▶ **α -sublevel set** of $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is
 $C_\alpha = \{x \in \text{dom } f \mid f(x) \leq \alpha\}$
- ▶ sublevel sets of convex functions are convex sets (but converse is false)
- ▶ **epigraph** of $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is
 $\text{epi } f = \{(x, t) \in \mathbf{R}^{n+1} \mid x \in \text{dom } f, f(x) \leq t\}$



- ▶ f is convex if and only if $\text{epi } f$ is a convex set

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Example: log-normal random variable

- ▶ suppose $X \sim \mathcal{N}(\mu, \sigma^2)$
- ▶ with $f(u) = \exp u$, $Y = f(X)$ is log-normal
- ▶ we have $\mathbf{E}f(X) = \exp(\mu + \sigma^2/2)$
- ▶ Jensen's inequality is

$$f(\mathbf{E}X) = \exp \mu \leq \mathbf{E}f(X) = \exp(\mu + \sigma^2/2)$$

which indeed holds since $\exp \sigma^2/2 > 1$

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Jensen's inequality

- ▶ **basic inequality**: if f is convex, then for
 $x, y \in \text{dom } f, 0 \leq \theta \leq 1$,

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

- ▶ **extension**: if f is convex and z is a random variable on $\text{dom } f$,

$$f(\mathbf{E}z) \leq \mathbf{E}f(z)$$

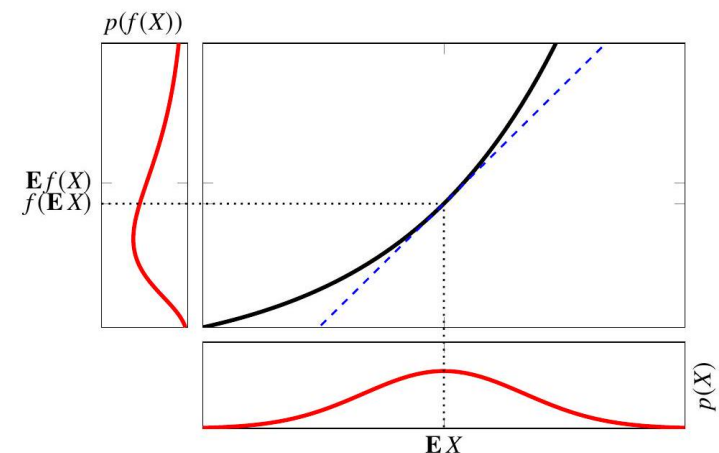
- ▶ basic inequality is special case with discrete distribution

$$\text{prob}(z = x) = \theta, \quad \text{prob}(z = y) = 1 - \theta$$

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Example: log-normal random variable



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Outline

Convex functions

Operations that preserve convexity

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Nonnegative scaling, sum, and integral

- ▶ **nonnegative multiple:** αf is convex if f is convex, $\alpha \geq 0$
- ▶ **sum:** $f_1 + f_2$ convex if f_1, f_2 convex
- ▶ **infinite sum:** if $f_1, f_2, \dots$ are convex functions, infinite sum $\sum_{i=1}^{\infty} f_i$ is convex
- ▶ **integral:** if $f(x, \alpha)$ is convex in x for each $\alpha \in \mathcal{A}$, then $\int_{\alpha \in \mathcal{A}} f(x, \alpha) d\alpha$ is convex
- ▶ there are analogous rules for concave functions

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Showing a function is convex

methods for establishing convexity of a function f

1. verify definition (often simplified by restricting to a line)
2. for twice differentiable functions, show $\nabla^2 f(x) \geq 0$
 - recommended only for **very simple** functions
3. show that f is obtained from simple convex functions by operations that preserve convexity
 - nonnegative weighted sum
 - composition with affine function
 - pointwise maximum and supremum
 - composition
 - minimization
 - perspective

you'll mostly use methods 2 and 3

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Composition with affine function

(pre-)composition with affine function: $f(Ax + b)$ is convex if f is convex

examples

- ▶ log barrier for linear inequalities

$$f(x) = -\sum_{i=1}^m \log(b_i - a_i^T x),$$

$$\text{dom } f = \{x \mid a_i^T x < b_i, i = 1, \dots, m\}$$

- ▶ norm approximation error: $f(x) = \|Ax - b\|$ (any norm)

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Pointwise maximum

if $f_1, \dots, f_m$ are convex, then $f(x) = \max \{f_1(x), \dots, f_m(x)\}$ is convex

examples:

- ▶ piecewise-linear function: $f(x) = \max_{i=1,\dots,m} (a_i^T x + b_i)$
- ▶ sum of r largest components of $x \in \mathbf{R}^n$:

$$f(x) = x_{[1]} + x_{[2]} + \cdots + x_{[r]}$$

($x_{[i]}$ is i th largest component of x)

Proof:

$$f(x) = \max \{x_{i_1} + x_{i_2} + \cdots + x_{i_r} \mid 1 \leq i_1 < i_2 < \cdots < i_r \leq n\}$$

(all the possible combinations of r different components of x)

Partial minimization

the function $g(x) = \inf_{y \in C} f(x, y)$ is called the **partial minimization** of f (w.r.t. y)

- ▶ if $f(x, y)$ is convex in (x, y) and C is a convex set, then partial minimization g is convex

Pf: special case when there exists a y_x such that $g(x) = f(x, y_x)$

examples

- $f(x, y) = x^T A x + 2x^T B y + y^T C y$ with

$$\begin{bmatrix} A & B \\ B^T & C \end{bmatrix} \geq 0, \quad C > 0$$

minimizing over y gives

$$g(x) = \inf_y f(x, y) = x^T (A - BC^{-1}B^T) x$$

g is convex, hence Schur complement $A - BC^{-1}B^T \geq 0$

- ▶ distance to a set: $\text{dist}(x, S) = \inf_{y \in S} \|x - y\|$ is convex if S is convex.

$$(x - y = (I, -I)??.)$$

Pointwise supremum

if $f(x, y)$ is convex in x for each $y \in \mathcal{A}$, then $g(x) = \sup_{y \in \mathcal{A}} f(x, y)$ is convex

examples

- ▶ distance to farthest point in a set C : $f(x) = \sup_{y \in C} \|x - y\|$
- ▶ maximum eigenvalue of symmetric matrix: for $X \in \mathbf{S}^n$, $\lambda_{\max}(X) = \sup_{\|y\|_2=1} y^T X y$ is convex
- ▶ support function of a set C : $S_C(x) = \sup_{y \in C} y^T x$ is convex

Composition with scalar functions

- ▶ composition of $g : \mathbf{R}^n \rightarrow \mathbf{R}$ and $h : \mathbf{R} \rightarrow \mathbf{R}$ is $f(x) = h(g(x))$
(written as $f = h \circ g$)
- ▶ composition f is convex if
 - g convex, h convex, $\tilde{h}$ nondecreasing
 - or g concave, h convex, $\tilde{h}$ nonincreasing
 (monotonicity must hold for extended-value extension $\tilde{h}$)
- ▶ proof (for $n = 1$, differentiable g, h)

$$f''(x) = h''(g(x))g'(x)^2 + h'(g(x))g''(x)$$

examples

- ▶ $f(x) = \exp g(x)$ is convex if g is convex
- ▶ $f(x) = 1/g(x)$ is convex if g is concave and positive

General composition rule

- ▶ composition of $g : \mathbf{R}^n \rightarrow \mathbf{R}^k$ and $h : \mathbf{R}^k \rightarrow \mathbf{R}$ is $f(x) = h(g(x)) = h(g_1(x), g_2(x), \dots, g_k(x))$
- ▶ f is convex if h is convex and for each i one of the following holds
 - g_i convex, $\tilde{h}$ nondecreasing in its i th argument
 - g_i concave, $\tilde{h}$ nonincreasing in its i th argument
 - g_i affine
- ▶ you will use this composition rule **constantly** throughout this course
- ▶ you need to commit this rule to memory

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Examples

- ▶ $\log \sum_{i=1}^m \exp g_i(x)$ is convex if g_i are convex
- ▶ $f(x) = p(x)^2/q(x)$ is convex if
 - p is nonnegative and convex
 - q is positive and concave(Noting x^2/t is convex of t, x).
- ▶ composition rule subsumes others, e.g.,
 - αf is convex if f is, and $\alpha \geq 0$
 - sum of convex (concave) functions is convex (concave)
 - max of convex functions is convex
 - min of concave functions is concave

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Perspective

- ▶ the **perspective** of a function $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is the function $g : \mathbf{R}^n \times \mathbf{R} \rightarrow \mathbf{R}$,

$$g(x, t) = tf(x/t), \quad \text{dom } g = \{(x, t) \mid x/t \in \text{dom } f, t > 0\}$$

- ▶ g is convex if f is convex

examples

- ▶ $f(x) = x^T x$ is convex; so $g(x, t) = x^T x/t$ is convex for $t > 0$
- ▶ $f(x) = -\log x$ is convex; so relative entropy $g(x, t) = t \log t - t \log x$ is convex on $\mathbf{R}_{++}^2$

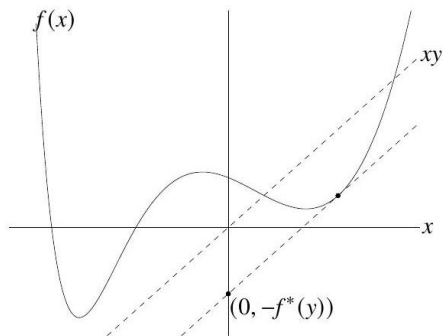
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Conjugate function

- the **conjugate** of a function f is

$$f^*(y) = \sup_{x \in \text{dom } f} (y^T x - f(x))$$



- f^* is convex (even if f is not)
- very important concept.

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Examples

- negative logarithm $f(x) = -\log x$

$$f^*(y) = \sup_{x > 0} (xy + \log x) = \begin{cases} -1 - \log(-y) & y < 0 \\ \infty & \text{otherwise} \end{cases}$$

- strictly convex quadratic, $f(x) = (1/2)x^T Qx$ with $Q \in \mathbf{S}_{++}^n$

$$f^*(y) = \sup_x (y^T x - (1/2)x^T Qx) = \frac{1}{2}y^T Q^{-1}y$$

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Basic Properties

Fenchel Inequality

$$f(x) + f^*(y) \geq x^T y$$

Examples:

- $f(x) = \frac{1}{2}x^T Qx$ where $Q \in \mathbf{S}_{++}^n$.

$$x^T y \leq \frac{1}{2}x^T Qx + \frac{1}{2}x^T Q^{-1}x$$

- $1/p + 1/q = 1$, $p, q > 1$, $a, b \in \mathbb{R}$

$$ab \leq |a|^p/p + |b|^q/q$$

- $f(x) = a^T x + b, x \in \mathbb{R}^n$.

$$f^*(y) = \begin{cases} -b & y = a \\ +\infty & \text{otherwise} \end{cases}$$

$$f^{**}(x) = a^T x + b \text{ for } x \in \mathbb{R}^n$$

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Conjugate of the Conjugate

If $f(x)$ is convex and $\text{dom}(f) = \mathbb{R}^n$, Then

$$f^{**} = f$$

outline of pf:

- 1: $f(x) = \sup\{g(x) | g \text{ affine}, g(z) \leq f(z) \text{ for all } z\}$
- 2: If $f(x) \geq g(x)$ for all x then $f^*(y) \leq g^*(y)$ for all y (by def) then $f^{**}(x) \geq g^{**}(x)$ for all x .
- 3: 1+2 $\Rightarrow$
 $f^{**}(x) \geq \sup\{g(x) | g \text{ affine}, g(z) \leq f(z) \text{ for all } z\} = f(x)$.
4. $f(x) \geq f^{**}(x)$
 (why? noting that $f(x) \geq x^T y - f^*(y)$ for all $y \Rightarrow f(x) \geq f^{**}(x) = \sup_y (x^T y - f^*(y))$)

1+4 $\Rightarrow f^{**}(x) = f(x)$ for all $x \in \mathbb{R}^n$.

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Suppose f is convex and differentiable with $\text{dom}(f) = \mathbf{R}^n$.
 $y^* = \nabla f(x^*)$ for any x^* . Then:

$$f^*(y^*) = x^{*T} y^* - f(x^*)$$

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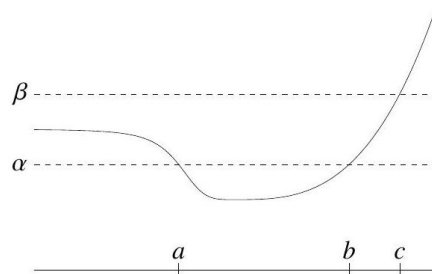


Quasiconvex functions

- $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is **quasiconvex** if **dom** f is convex and the sublevel sets

$$S_\alpha = \{x \in \text{dom } f \mid f(x) \leq \alpha\}$$

are convex for all α



- f is **quasiconcave** if $-f$ is quasiconvex
- f is **quasilinear** if it is quasiconvex and quasiconcave

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Examples

- $\sqrt{|x|}$ is quasiconvex on $\mathbf{R}$
- $\text{ceil}(x) = \inf\{z \in \mathbf{Z} \mid z \geq x\}$ is quasilinear ($\mathbf{Z}$: the set of integers)
- $\log x$ is quasilinear on $\mathbf{R}_{++}$
- $f(x_1, x_2) = x_1 x_2$ is quasiconcave on $\mathbf{R}_{++}^2$
- linear-fractional function

$$f(x) = \frac{a^T x + b}{c^T x + d}, \quad \text{dom } f = \{x \mid c^T x + d > 0\}$$

is quasilinear

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Example: Internal rate of return

- ▶ cash flow $x = (x_0, \dots, x_n)$; x_i is payment in period i (to us if $x_i > 0$)
- ▶ we assume $x_0 < 0$ (i.e., an initial investment) and $x_0 + x_1 + \dots + x_n > 0$
- ▶ **net present value (NPV)** of cash flow x , for interest rate r , is $PV(x, r) = \sum_{i=0}^n (1+r)^{-i} x_i$
- ▶ **internal rate of return (IRR)** is smallest interest rate for which $PV(x, r) = 0$:

$$\text{IRR}(x) = \inf\{r \geq 0 \mid \text{PV}(x, r) = 0\}$$

- ▶ IRR is quasiconcave: superlevel set is intersection of open halfspaces

$$\text{IRR}(x) \geq R \iff \sum_{i=0}^n (1+r)^{-i} x_i > 0 \text{ for } 0 \leq r < R$$

Properties of quasiconvex functions

- ▶ **modified Jensen inequality:** for quasiconvex f

$$0 \leq \theta \leq 1 \implies f(\theta x + (1 - \theta)y) \leq \max\{f(x), f(y)\}$$

How to prove it?

- **first-order condition:** differentiable f with convex domain is quasiconvex if and only if

$$f(y) \leq f(x) \implies \nabla f(x)^T (y - x) \leq 0$$

- ▶ **sum** of quasiconvex functions is not necessarily quasiconvex

Outline

Summary

- ▶ Convex (QuasiConvex) functions.
- ▶ How to prove a function is convex.
- ▶ Conjugate function and its properties.